LAB EXPERIMENT RECORD

Set 15: Product Sales Analysis

Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using Tableau
Name	Peram Mahitha
Register Number	192524136
Date	8 October 2026
Day	Thursday

Aim

To analyse monthly product sales using a grouped bar chart, a stacked area chart and a table, and to build an interactive Tableau dashboard for exploring sales data.

Dataset: Monthly Product Sales

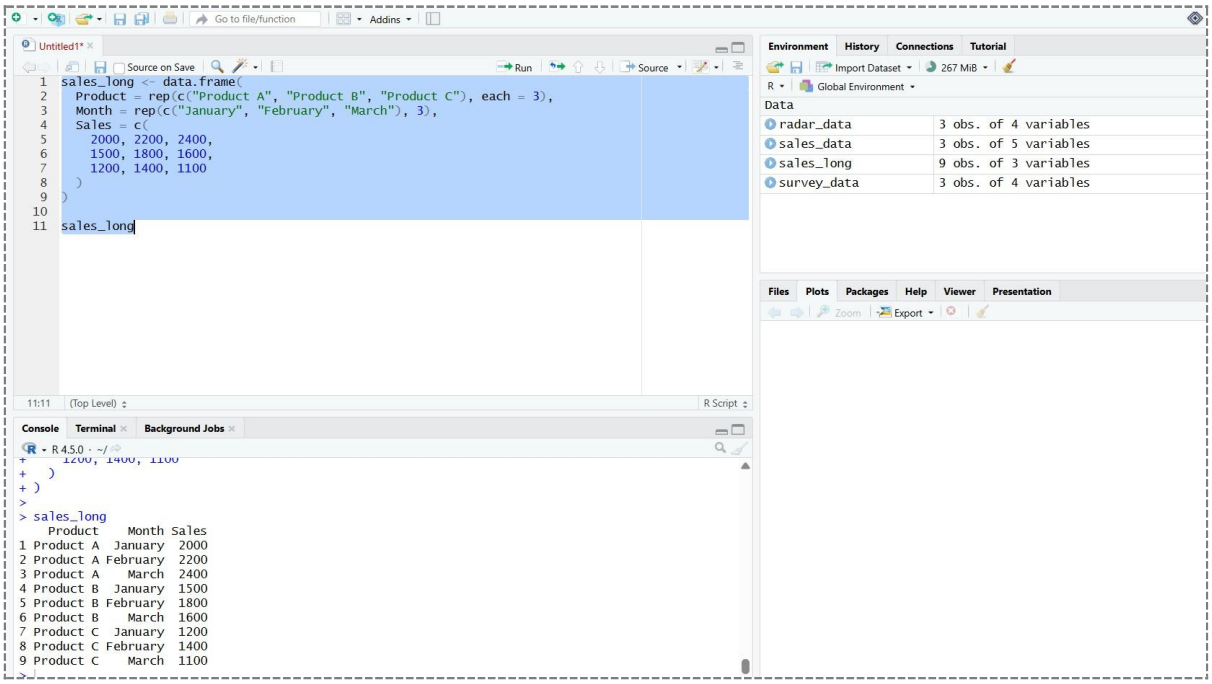
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart of First Quarter Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Steps in Tableau:

1. Connect to the sales file. Select January, February and March Sales, right-click and choose Pivot to get the fields Pivot Field Names (rename to Month) and Pivot Field Values (rename to Sales).
2. Create a new sheet named "Grouped Bar".
3. Drag Product Name to Columns, then drag Month to Columns next to it so the bars are grouped by product.
4. Drag Sales to Rows (SUM) and Month to Color.
5. Show Sales values as labels, set the title "Product Sales - First Quarter", and label the axes "Product" and "Sales".



Output 1: Grouped Bar Chart of Product Sales (Q1)

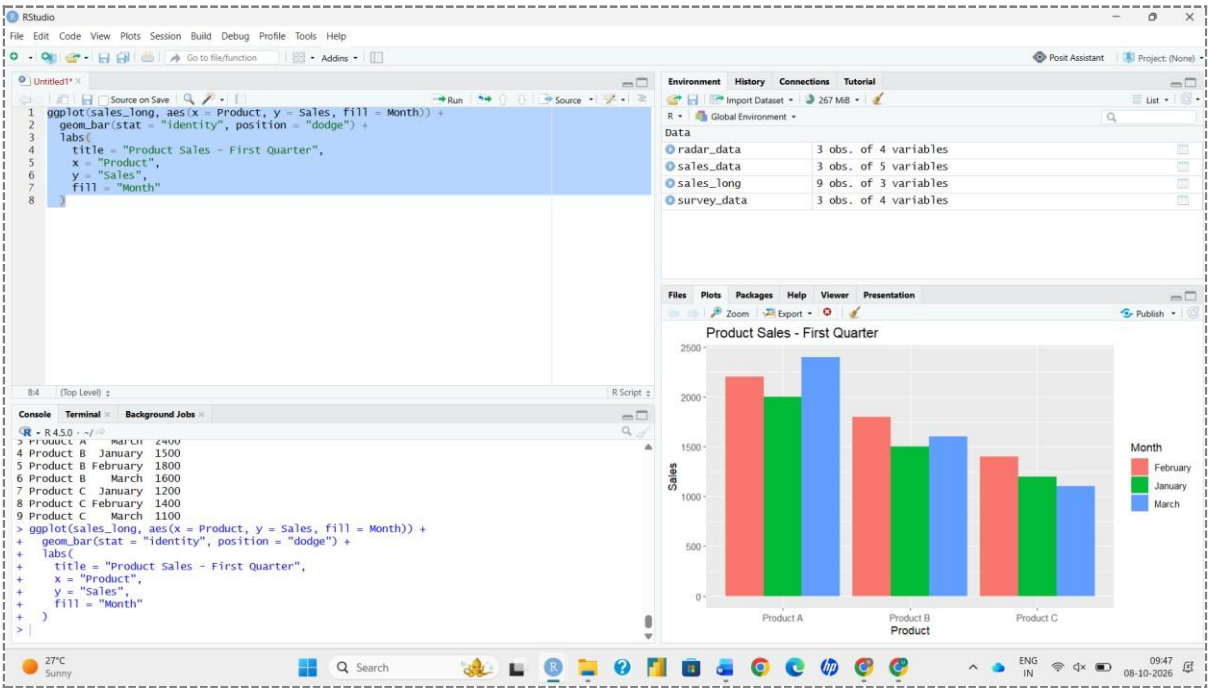
Observation: Product A has the highest sales in every month and grows steadily (2000, 2200, 2400). Products B and C both peak in February and fall in March.

PROGRAM 2: Stacked Area Chart of Sales Trend

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Steps in Tableau:

1. Add a Month Number field (January = 1, February = 2, March = 3) using a calculated field so that the month axis is continuous.
2. Create a new sheet named "Stacked Area".
3. Drag Month Number to Columns and set it to Continuous; drag Sales to Rows (SUM).
4. Drag Product Name to Color and change the mark type to Area; Tableau stacks the areas automatically.
5. Edit the axis labels to show month names, add the title "Sales Trend by Product" and show the legend.



Output 2: Stacked Area Chart of Overall Sales Trend

Observation: Total sales rise from 4700 in January to 5400 in February and then dip to 5100 in March. Product A contributes the largest share of the total (about 43% of the quarter's 15200).

PROGRAM 3: Table of Monthly Sales Data

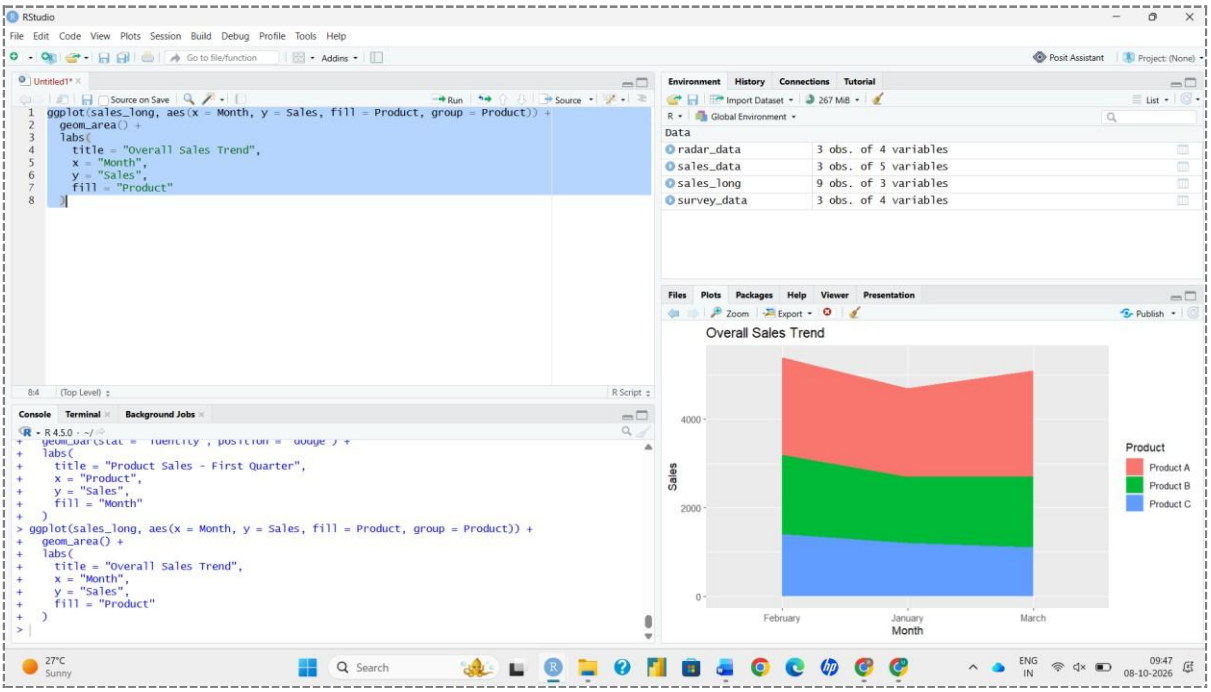
Build a table to show the monthly sales data for each product. Label the table elements.

Steps in Tableau:

1. Create a new sheet named "Sales Table".
2. Drag Product Name to Rows and Month to Columns.
3. Drag Sales to Text.
4. Use Analytics > Totals (or Worksheet > Show Totals) to add row and column grand totals.
5. Add the title "Monthly Sales by Product" and format the values with thousand separators.

Expected totals (for verification):

Product	January	February	March	Total
Product A	2000	2200	2400	6600
Product B	1500	1800	1600	4900
Product C	1200	1400	1100	3700
Total	4700	5400	5100	15200



Output 3: Table of Monthly Product Sales

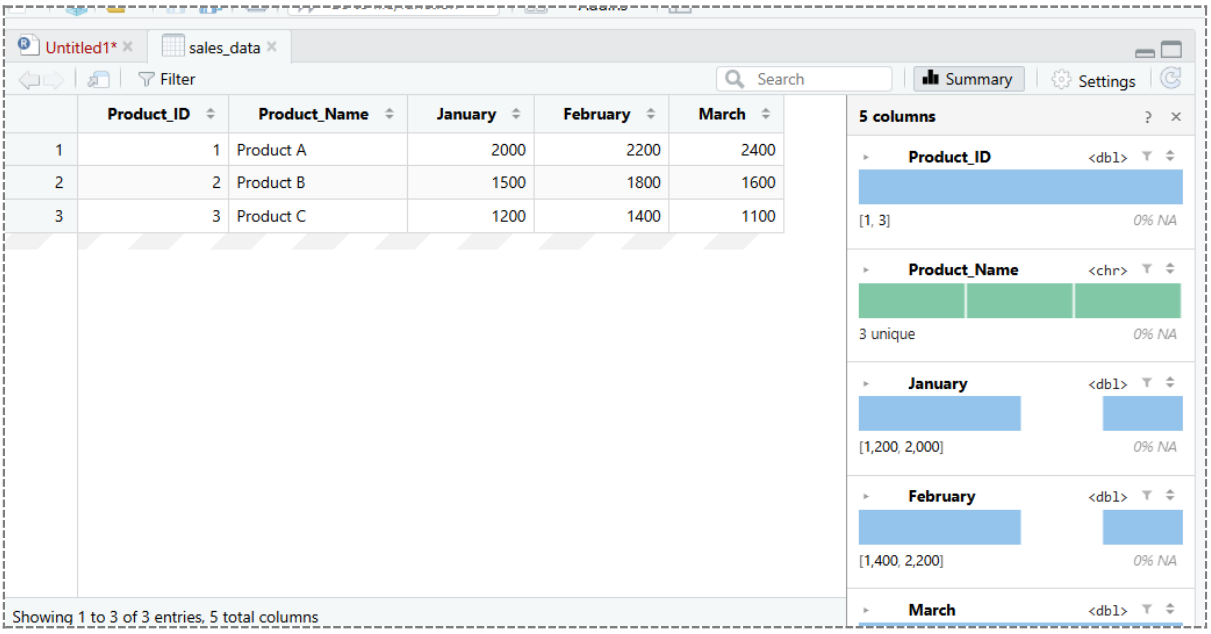
Observation: The table gives exact monthly figures; Product A is the top seller with 6600 units of sales in the quarter, followed by Product B (4900) and Product C (3700).

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart and table for interactive exploration of sales data.

Steps in Tableau:

1. Click New Dashboard at the bottom of the workbook and rename it.
2. Set Size to Automatic (or Fixed: Desktop Browser) from the Dashboard pane.
3. Drag Grouped Bar, Stacked Area and Sales Table from the Sheets list onto the dashboard canvas.
4. Add a Title (Dashboard > Show Title) and arrange the sheets using Tiled layout containers.
5. Select the first chart and click Use as Filter so that selecting a product in the bar chart filters the area chart and the table.
6. Use Dashboard > Actions to add a Highlight action, and enable Show Filters on the sheets for dropdown filtering.
7. Click different marks to test interactivity and confirm all views update together.



Output 4: Interactive Product Sales Dashboard

Observation: Selecting a product highlights its bars, its area in the trend chart and its row in the table, so product performance can be compared across months quickly.

Procedure

1. Connect the product sales data to Tableau and pivot the three monthly columns into Month and Sales.
2. Build the grouped bar chart with Product Name and Month on Columns and SUM(Sales) on Rows.
3. Build the stacked area chart with Month Number on Columns, SUM(Sales) on Rows and Product Name on Color.
4. Build the table with Product Name on Rows, Month on Columns and Sales as text.
5. Combine the three sheets in a dashboard and add filter and highlight actions.
6. Interact with the dashboard to explore sales by product and month.

Result

The monthly product sales data was successfully visualized using a grouped bar chart, stacked area chart and table, and an interactive Tableau dashboard was created for exploring sales data.